

WELELE MEDIA™

Final Architecture Decisions & Immediate Actions

Digital IP Engine — Architecture Evolution Decision

Final engineering position: Welele is not being rebuilt; it is being promoted from a Content Platform MVP into a Digital IP Engine.

1. Strategic Architecture — LOCK

Welele is not fundamentally a streaming app. The target architecture is a **Digital IP Engine + Creator Operating System + Experience Engine + Audience Intelligence**. The streaming interface is the audience surface.

Digital IP → Story → Creation → Production → Distribution → Audience → Data → Intelligence → Better Stories → More Valuable IP

2. PostgreSQL / Supabase — Single Source of Truth

Retire the JSON/in-memory store as production authority. Do not maintain dual authority. Target flow: React → FastAPI → Domain Services → PostgreSQL. Introduce a repository/data-access layer and migrate domain by domain rather than performing a single giant rewrite.

3. Establish the IP Spine First

Current model: Creator → Story → Episode → Video. Target model: **IP → Story World / Characters / Rights / Contributors / Story Packages → Series → Seasons → Episodes → Media → Economics**. Everything new should attach to this spine. Do not build more orphaned features.

4. Story Forge — Canonical Story Package

Story Forge must produce a persisted, versioned Story Package rather than ephemeral text. It should include IP, concept, logline, story bible, characters, relationships, world, timeline, language/dialect, episode architecture, beats, dialogue, hook strategy, cliffhanger strategy, continuity rules and production notes. Track model, version, human edits and production lineage.

5. AI — Efficiency Layer, Not the Moat

Keep Gemini and future models behind a Model Gateway. Welele owns the canonical schema, workflow, evaluation, prompts/versioning, story history and performance intelligence. The model is replaceable; the accumulated intelligence is not.

6. Financial Architecture — Correct Before Real Money

The current wallet abstraction is useful, but it should not be described as true double-entry accounting. Build a Financial Transaction → Journal → Journal Lines model with balanced debits and credits. Use idempotency and database-level atomicity. Wallet balances may be materialized, but the journal is the financial truth.

7. Event Spine — Build Early

Establish canonical telemetry for sessions, impressions, title opens, episode starts, playback progress, cliffhanger reached, paywall exposure, unlocks, completion, next-episode starts, shares and referrals. Events must carry context such as experience/version, surface, traffic source, campaign, experiment, device, network and language.

8. Events → Analytics → Intelligence

Keep three distinct layers: **Events** answer what happened; **Analytics** answer what it means; **Intelligence** answers what should we do. Intelligence must feed Story Forge, Creator Studio, WEE, commissioning, distribution and IP valuation.

9. WEE — Experience Decision Engine

Keep the existing Experience Engine and evolve it. WEE should move from controlling presentation to making evidence-informed experience decisions using audience behaviour, content performance, experiments and intelligence. The target loop is: WEE decision → Viewer behaviour → Telemetry → Intelligence → Better WEE decision.

10. Rights — Treat Them as an Asset System

Rights must include holder, right type, territory, language, medium, exclusivity, start/end dates, contract, percentage and status. This enables future licensing, adaptation, remakes and commercial rights decisions.

11. Media — Asset Pipeline, Not Just video_url

Move from Episode → video_url toward Episode → Media Asset family: master, HLS/renditions, thumbnails, captions, subtitles and localisations. Target lifecycle: Upload → Processing → Transcoding → Preflight → Moderation → Approved → Publish. Actual transcoding infrastructure can follow once the data architecture is correct.

12. Payments — Financial Correctness Before Provider Breadth

Do not rush multiple live payment integrations. First establish Purchase → Payment Intent → Provider → Webhook → Verification → Ledger Transaction → Coins → Unlock, with idempotency and reconciliation. Then replace sandbox adapters with production providers.

13. Predictive Valuation — P3, Not Now

Do not build speculative AI valuation before trustworthy evidence exists. First collect audience, revenue, retention, continuation, production cost, rights, creator performance, market, language and distribution data. Valuation follows evidence.

14. Architecture Style — Stay Modular

Do not turn the platform into microservices yet. Keep a modular domain platform inside FastAPI with clear modules for IP, Story, Creator, Content, Media, Audience, Experience, Commerce, Rights, Telemetry and Intelligence. Split services only when there is an actual scaling or operational reason.

15. Final Implementation Order

Gate	Primary outcome
0 — Architecture Contract	Freeze domain model, terminology, IDs, ownership, event envelope, rights, financial model and boundaries
1 — Data Truth	PostgreSQL authority; repositories; IP → Series → Episode → Media.
2 — Economic Truth	Wallet, financial journal, transactions, entitlements, rights splits, idempotency.
3 — Story Truth	Story Forge → versioned Story Package → IP → Creator Studio.
4 — Behavioural Truth	Viewer → canonical telemetry → analytics.
5 — Experience Intelligence	Analytics → Story Intelligence → WEE.
6 — Production Infrastructure	R2 → transcoder → HLS → preflight → moderation.
7 — Intelligence	Story, Creator, Audience, Experience Intelligence and IP Economics.

16. Immediate Actions

- **Freeze the architecture contract.** No new major feature should bypass the agreed domain spine.
- **Complete the remaining vertical-slice audit** before committing to detailed engineering durations.
- **Design the canonical IP, Rights, Story Package, Event and Financial models.**
- **Introduce the persistence abstraction** so JSON can be removed without rewriting every route at once.
- **Prioritise P0 truth:** PostgreSQL authority, IP identity/rights and atomic financial transactions.
- **Establish telemetry early** so subsequent product decisions are based on real behaviour.
- **Connect Story Forge to persisted IP and Creator Studio.**
- **Evolve WEE toward intelligence-driven decisioning** after the underlying event and analytics spine is trustworthy.
- **Keep implementation modular.** Avoid premature microservices and speculative AI infrastructure.

17. What We Are NOT Doing

- Not rebuilding Welele from scratch.
- Not making Gemini the strategic moat.
- Not maintaining JSON and PostgreSQL as competing authorities.
- Not calling the current wallet ledger true double-entry accounting.
- Not building predictive valuation before trustworthy evidence exists.
- Not introducing microservices merely for architectural fashion.
- Not expanding payment providers before transaction correctness is proven.

18. Final Engineering Principle

Make the underlying machine truthful. The existing prototype has already proved the major product surfaces and abstractions. The next step is to make IP, Rights, Story Packages, Events, Economics and Intelligence first-class persistent domains. Once those become architecture rather than presentation language, the Welele moat becomes real.

Final question for every future architecture decision:

“Does the Welele we have actually built contain the technical spine required to become the African Digital IP Engine we have decided to build?”